

PROJECT				JOB NO.			
CONTRACTOR :							
SUB-CONTRACTOR : Winner Engineering Pte Ltd							
TRADE		: ACMV SERVICES			DRAWING NO :		
FUNCTIONAL- TESTING COMMISSIONING CHECKLIST (PUMP)							
Unit Designation :							
Unit Make :							
Model no.							
Location :							
Serial no :							
S/N	Section	Inspection / Test Item	CHECKED CONDITION (Please tick inside checkbox)			Remarks	
			Yes	No	NA		
1	General	Check that the equipment is clean, free of debris, and undamaged before commencing operations.					
2	Documents	Verify chiller tag, make, model, serial number, refrigerant and rated capacity.					
3	Electrical	Verify supply voltage, phase sequence and running current on applicable phases.					
4	Controls	Test local start/stop, Auto/Manual, Remote/BMS and run/trip indications where provided.					
5	BMS	Compare BMS commands, statuses and analog readings with actual field conditions.					
6	Rotation	Verify motor/pump rotation by manufacturer-approved procedure after priming.					
7	Starting	Start at approved valve positions and speed; confirm immediate liquid delivery.					
8	Leakage	Check casing, flanges, seals/glands and seal flush during operation.					
9	Mechanical	Check noise, vibration and bearing temperatures.					
10	Flow	Measure pump water flow at the specified operating duty.					
11	Head	Record suction/discharge pressures and determine pump differential head.					
12	Motor loading	Record operating speed/frequency, current and input power where required.					
13	VSD	Test speed variation and differential-pressure control where fitted.					
14	Minimum flow	Verify minimum-flow/bypass response for pump and connected chiller where required.					
15	Valves	Test motorized valve command, travel, feedback and pump interlocks where fitted.					
16	Non-return valve	Verify check-valve operation on pump stop and standby operation.					
17	Duty/standby	Test lead/lag, duty/standby changeover and automatic standby start on failure where provided.					
18	Parallel pumps	Test simultaneous/parallel operation and staging where specified.					
19	Plant interlocks	Verify pump proof-of-flow and chiller enable/shutdown coordination.					
20	Faults	Test applicable overload, drive fault and low-flow/low-pressure alarms using approved simulation.					
21	Power recovery	Verify power-loss/restoration and restart behavior where specified.					
22	Balancing	Complete system balancing and verify final branch/plant flow requirements.					

23	Stopping	Verify normal stop and observe pressure transients.				
24	Stability	Run for specified test duration at agreed duties.				
25	Completion	Record final setpoints, control mode and operating settings; restore normal operation.				
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Comments to follow up Action :

Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.

CHECKED BY:		WITNESSED BY	INSPECTED BY
NAME			
DESIGNATION :			
COMPANY :			
SIGNATURE :			
DATE :			